



AIN SHAMS UNIVERSITY
FACULTY OF COMPUTER AND INFORMATION SCIENCES

A Proposed Research Plan for M.Sc. Degree
By

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Thesis Title

**Explainable Deep Learning and Large Language Models for
Cryptocurrency Forecasting**

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Research Are: (2) Data Science - Machine Learning
Applied Area: Cryptocurrencies
Sustainable Development: (8) Decent work and economic growth

Introduction

The financial markets have seen a big change with the emergence of cryptocurrencies, which have gained a lot of attention and are now considered a growing market that many investors want to enter. Cryptocurrencies suffer from significant price fluctuations, rising and falling due to factors like news, tweets, and trading volume. As a result, predicting their price has become complicated for investors.

With the huge amount of data available on cryptocurrencies, including their historical prices since they started, it has become possible to use machine learning techniques and language models to predict cryptocurrency prices.

Problem Definition

Cryptocurrencies are high-risk assets, and predicting their prices is challenging due to the influence of numerous factors, such as economic news, social media activity, daily transaction volumes, and the currency's historical data.

Despite advancements in analytical techniques, traditional models that rely solely on analyzing historical data have failed to provide accurate predictions of cryptocurrency prices. Therefore, it has become essential to develop a model that not only analyzes historical price data but also incorporates textual data from financial news, tweets, and posts on various platforms. Additionally, leveraging the indicators used by traders can help achieve the highest possible accuracy in predicting cryptocurrency prices.

In the process of forecasting cryptocurrency price, a model that works as a "black box" is not enough. The forecasting process needs to be transparent, and the reasons behind the generated results must be very clear so that investors can rely on these outcomes.

Therefore, the approach will also rely on explainable deep learning to create a transparent, reliable model that provides highly accurate and trustworthy results.

Research Objective

1. Developing an intelligent model capable of integrating textual data (Twitter tweets) with time-series data (historical prices and trading volumes).
2. Enhancing the accuracy of cryptocurrency price prediction using large language models based on explainable deep learning techniques.
3. Improving prediction accuracy by leveraging technical analysis methods employed by cryptocurrency traders.
4. Comparing the performance of the developed model with traditional methods for cryptocurrency price prediction.

Research Plan

5. Conducting a review of previous works in the field of cryptocurrency price prediction.
6. Collecting textual and historical data related to Ethereum.
7. Studying the technical analysis methods used by traders for predicting cryptocurrency prices.
8. Utilizing large language models and explainable deep learning networks to develop a model for cryptocurrency price prediction.
9. Test and evaluate the proposed model.
10. Compare results with old models.
11. Publish the research results.
12. Write the Thesis.

References

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